H MANJUNATHA

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PROFESSIONAL SUMMARY

Embedded C developer with hands-on experience in PIC18F4580 microcontrollers, firmware development, and CAN, UART, SPI, and I2C protocols. Developed embedded projects, including a 3-node CAN-based automotive dashboard, with focus on communication, peripheral interfacing, and debugging. Seeking a Firmware/Embedded Software Engineer role in a semiconductor or product-based engineering company.

TECHNICAL SKILLS

Languages: C, C++ (OOP Concepts Basics), Embedded C
Microcontrollers / Boards: PIC18F4580, GPIO, Timers, Interrupts, ADC, PWM
Communication Protocols: CAN, UART, SPI, I2C, TCP/UDP Sockets
Data Structures & Algorithms: Linked Lists, Stacks, Queues, Trees, Hash Tables, Sorting, Big-O Complexity Analysis
Tools & Platforms: MPLAB X IDE, MPLAB XC8, GCC, VS Code, Vim, Proteus, Ubuntu Linux / WSL, Blynk

KEY PROJECTS

CAN-Based Automotive Dashboard — Multi-Controller Vehicle Monitoring System

- Architected a 3-controller CAN bus network to display real-time RPM, speed, gear position, and indicator status on a car dashboard.
- Implemented interrupt-driven message handling to keep multi-node data synchronized with no observed message collision under test conditions.

Arbitrary Precision Calculator

- Implemented addition, subtraction, multiplication, division, modulus, and power operations on integers of unbounded size using a linked-list digit representation.
- Resolved carry/borrow propagation and pointer-management bugs to produce a stable, segmentfault-free arithmetic engine.

Image Steganography Tool — LSB Encoding/Decoding

- Developed an encoder/decoder pair that hides secret file data inside BMP pixel data using LSB substitution with zero visible image distortion.
- Preserved magic-string and metadata integrity across the encode/decode cycle to guarantee lossless data recovery.

MP3 Tag Reader and Editor

- Built a CLI tool to read and edit ID3v2 tags (Title, Artist, Album) in MP3 files using byte-level binary parsing.
- Resolved little-endian vs. big-endian byte-swapping issues and hardened the editor against buffer overflows during tag modification.

Inverted Search Engine

- Built a custom hash table with chaining to index text files and support fast keyword lookups by file and line number.
- Designed a memory-efficient inverted index (word → files → line positions) capable of parsing and tokenizing large text files.

Address Book Management System

- Developed a menu-driven CLI system to add, delete, modify, search, and sort contacts by name, phone, or email with persistent file storage.
- Implemented input validation to catch duplicate contacts and invalid data formats across sessions.

ACADEMIC PROJECTS

Automated Paralysis Patient Health Care System

- Developed an IoT-based healthcare system using Arduino Uno, accelerometer, DHT11, NodeMCU, 16×2 LCD, and Blynk to monitor patient movement and environmental conditions.
- Implemented gesture-based communication to convert patient hand movements into messages such as I Need Food, I Need Water, I Need Washroom, and Emergency, displayed on the LCD and transmitted through Blynk for remote monitoring.

EDUCATION

Bachelor of Engineering, (E&E) *2022 – 2025*
Jain Institute of Technology, Davangere (VTU) — 82.29%

Diploma (E&E) *2018 – 2021*
STJ Polytechnic, Harapanahalli —64%

Secondary (X), State Board *2018*
LHS High School, Hampasagara — 72%

CERTIFICATIONS & ACHIEVEMENTS

- Emertxe Certified Embedded Professional — Spot Award, Jan 2026
- Certified in Embedded C, Advanced C, Data Structures & Algorithms, C++ Programming, and Linux